Thread - 2024-12-24

https://x.com/RiikonenMarko/status/1871562225665925538

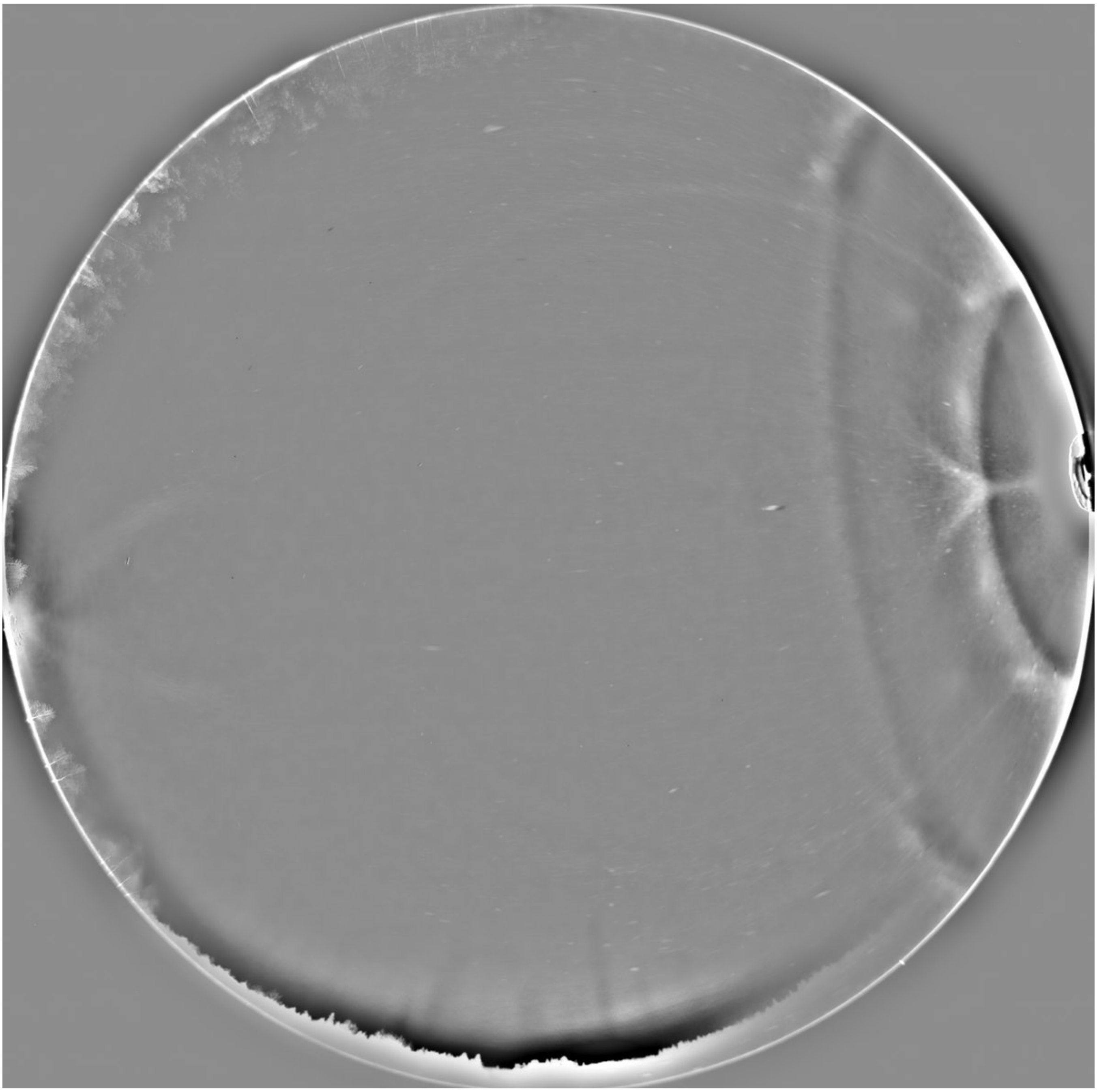
1/9 - 2024-12-24 14:23 UTC

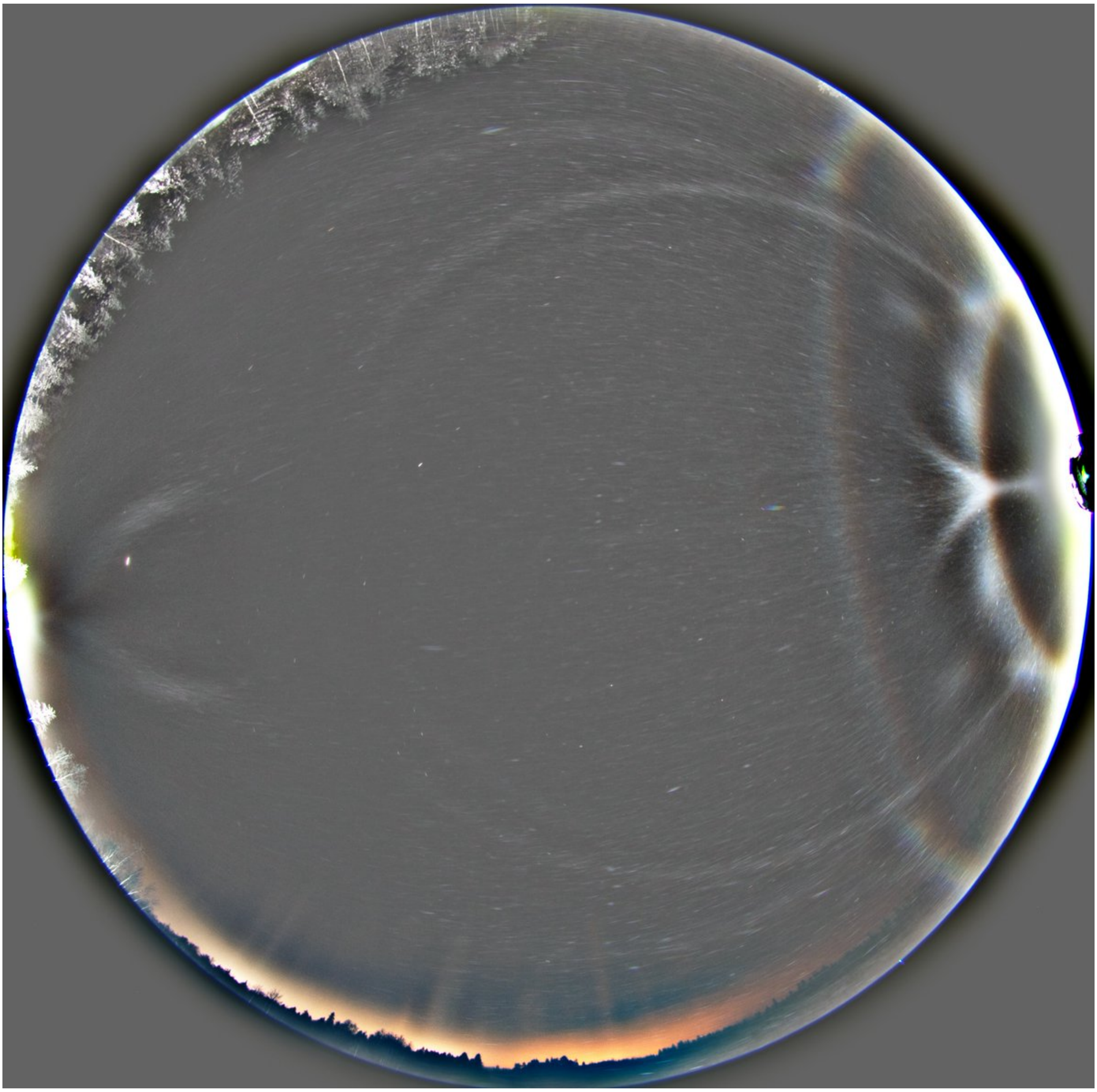
From last winter. Alternating two stacks, both of some five successive frames, separated by a couple of minutes of lesser action. As if the upper 35° plate is shape-shifting. Could 35° column arc be playing some role here? 10/11 November 2023, Rovaniemi, Toramo pit.

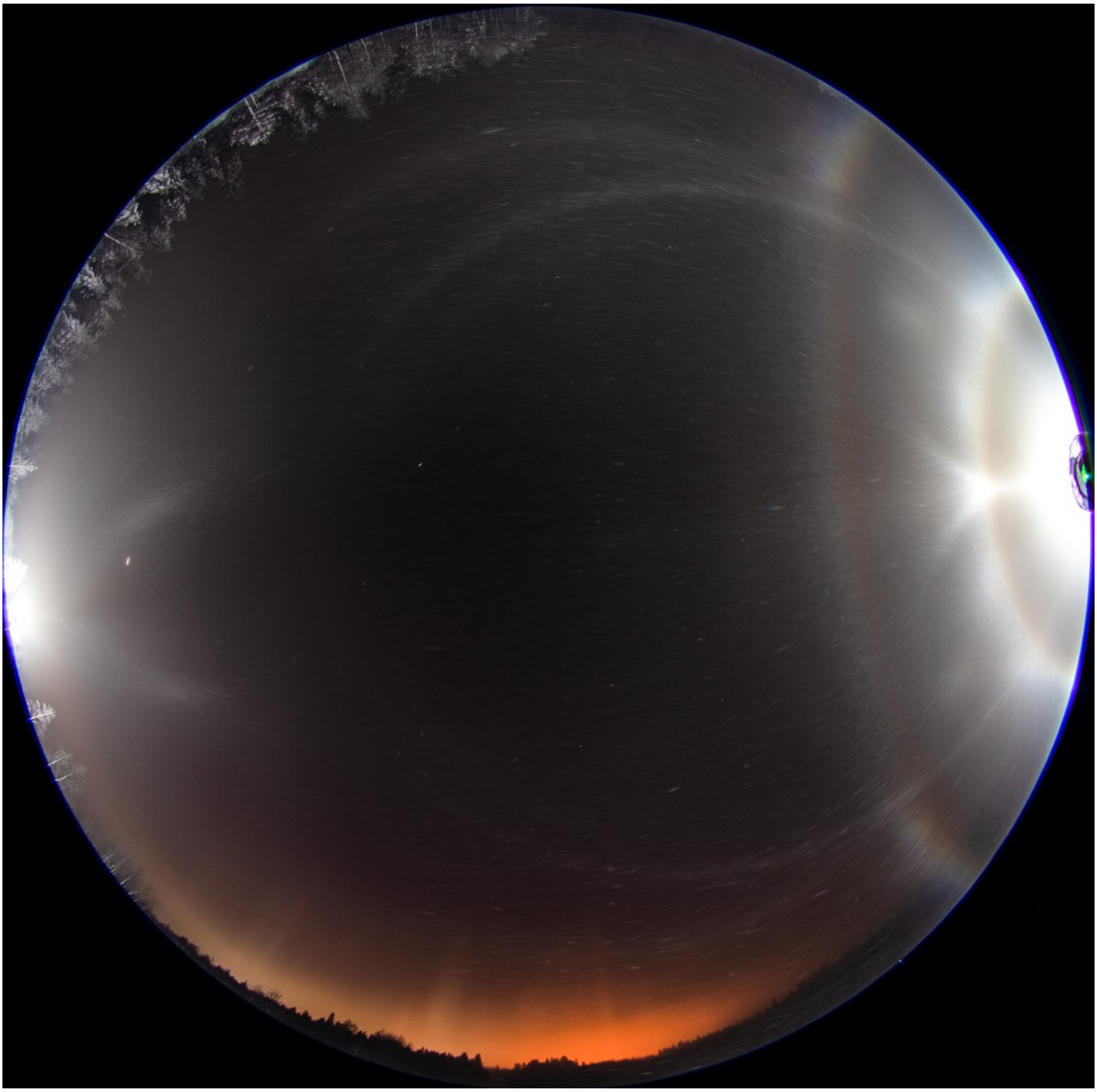
[video: 1871562225665925538-GfkVUOWXIAAWIYG.mp4]

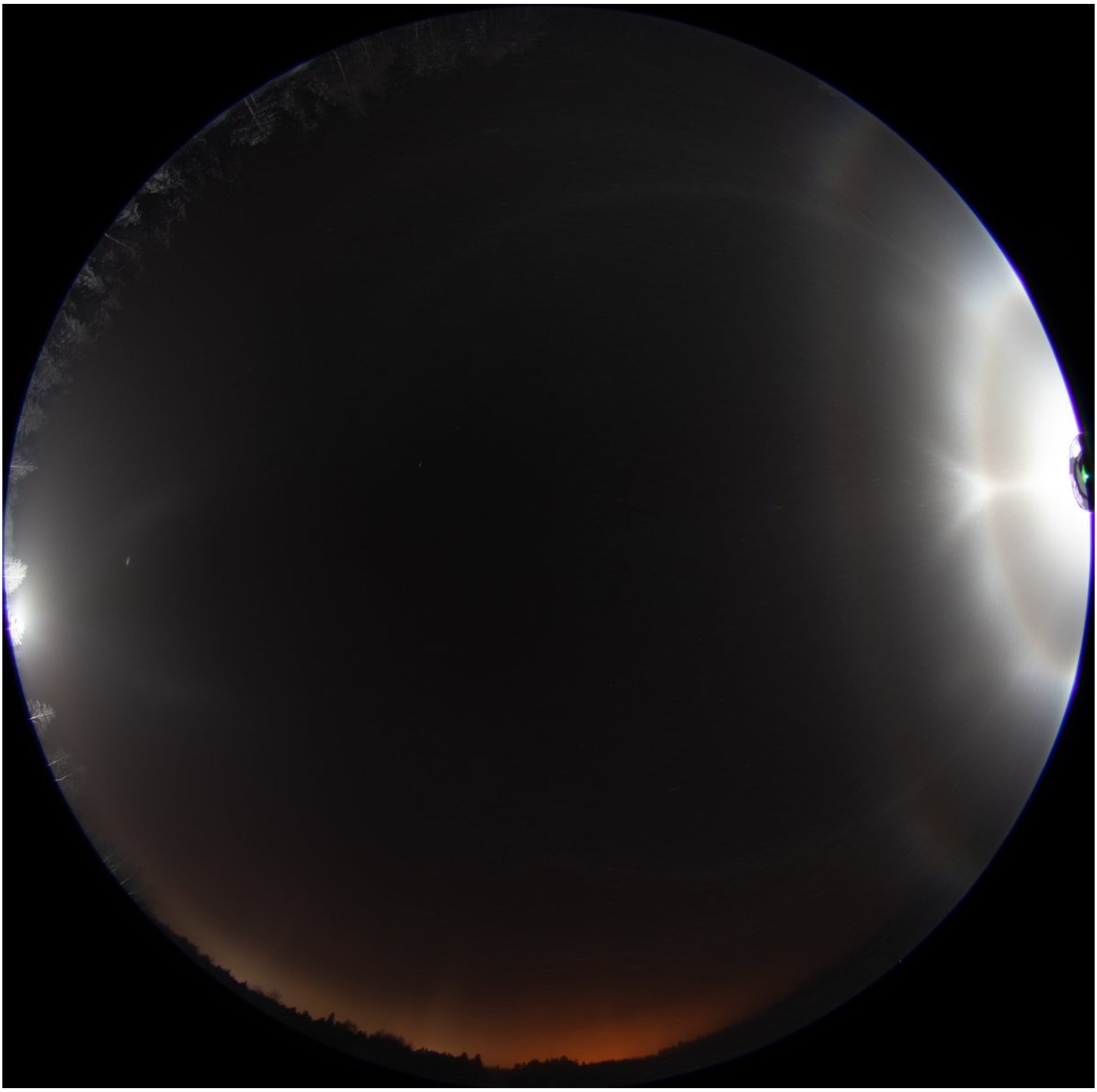
2/9 - 2024-12-24 17:59 UTC

10/11 November 2023, Rovaniemi. In this and next post down versions of the two stacks in the animation. Here frames 5275-5280.



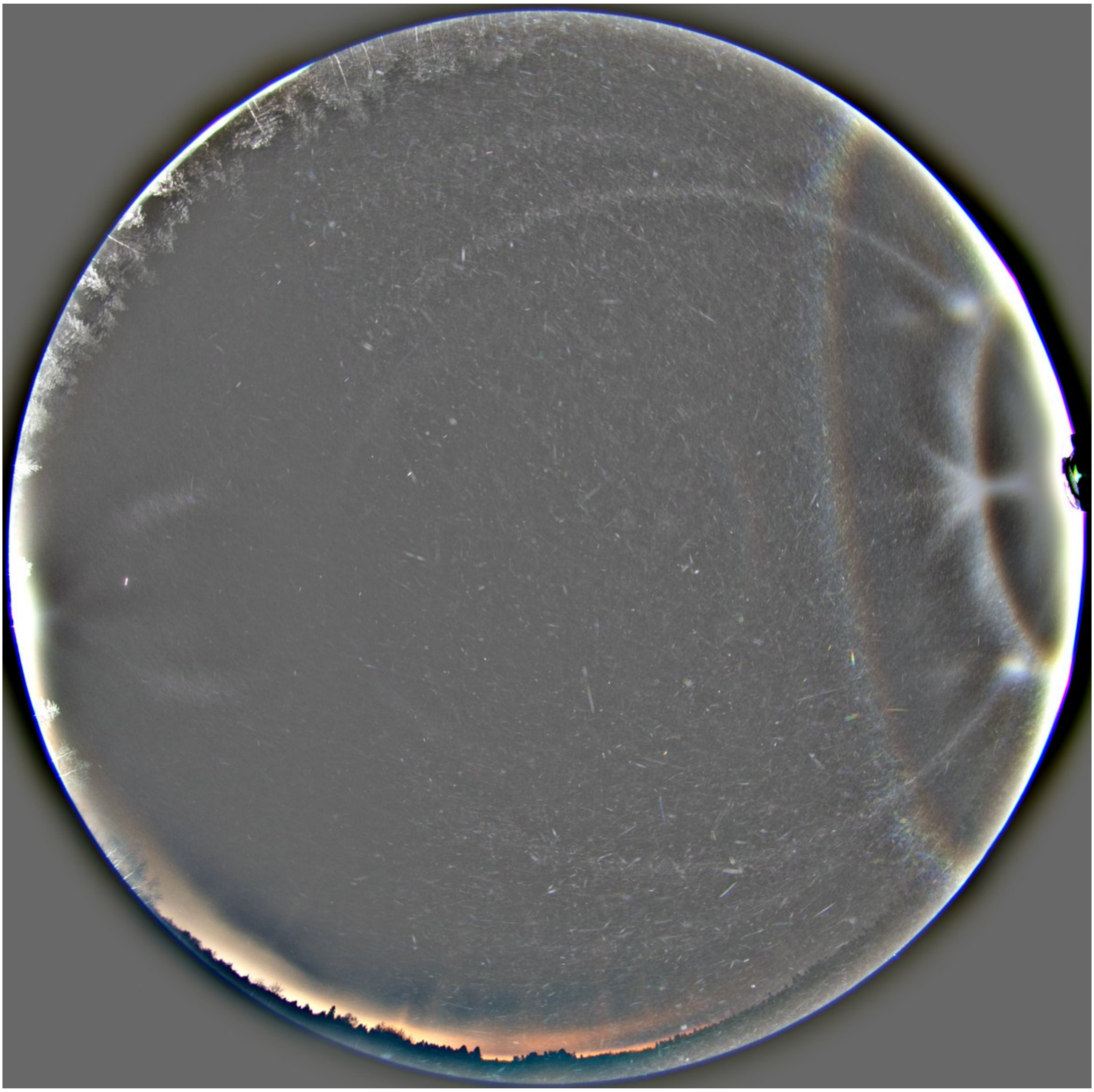


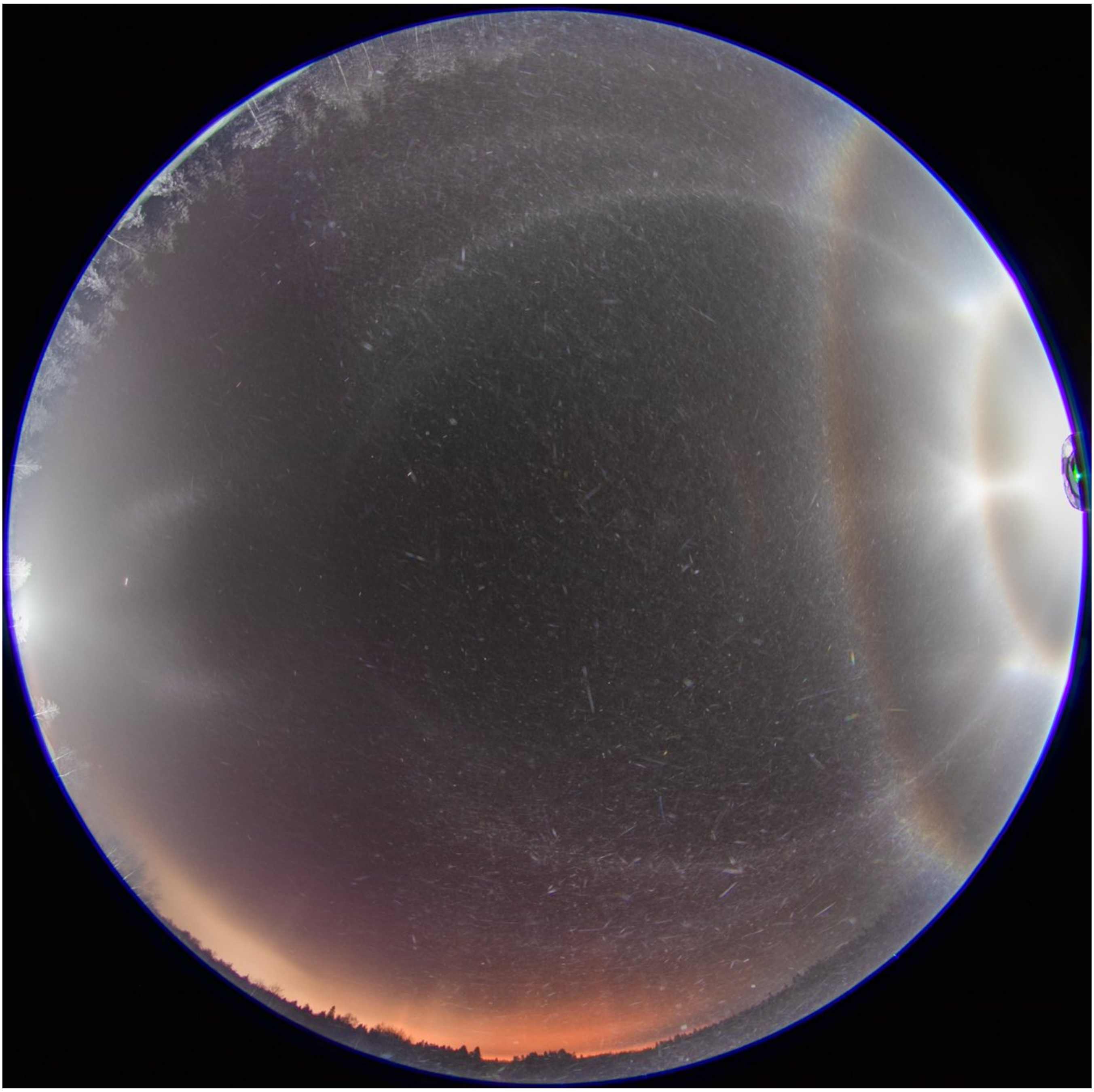


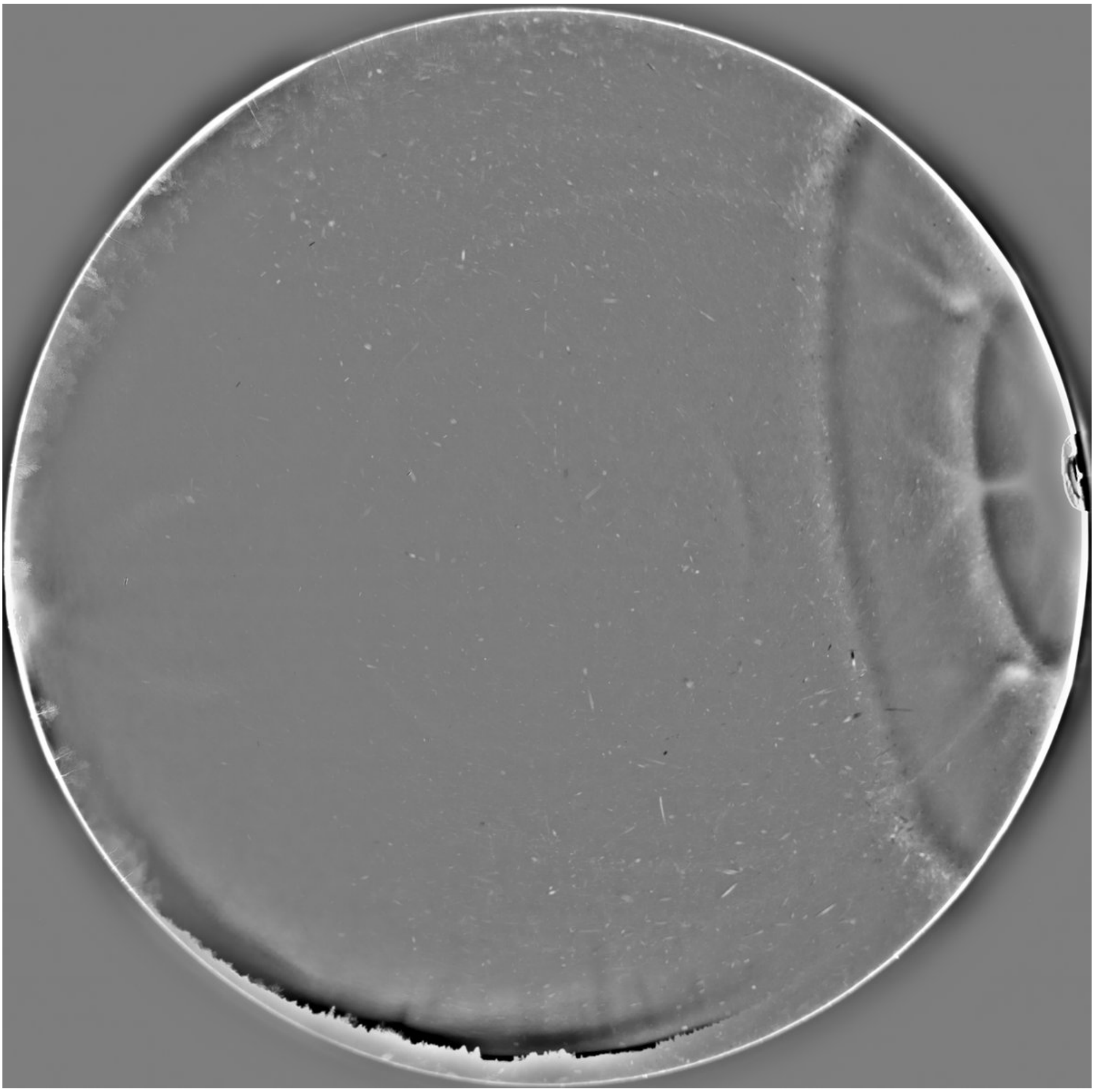


3/9 - 2024-12-24 18:01 UTC

10/11 November 2023, Rovaniemi. Frames 5295-5299. All 30s exposures.







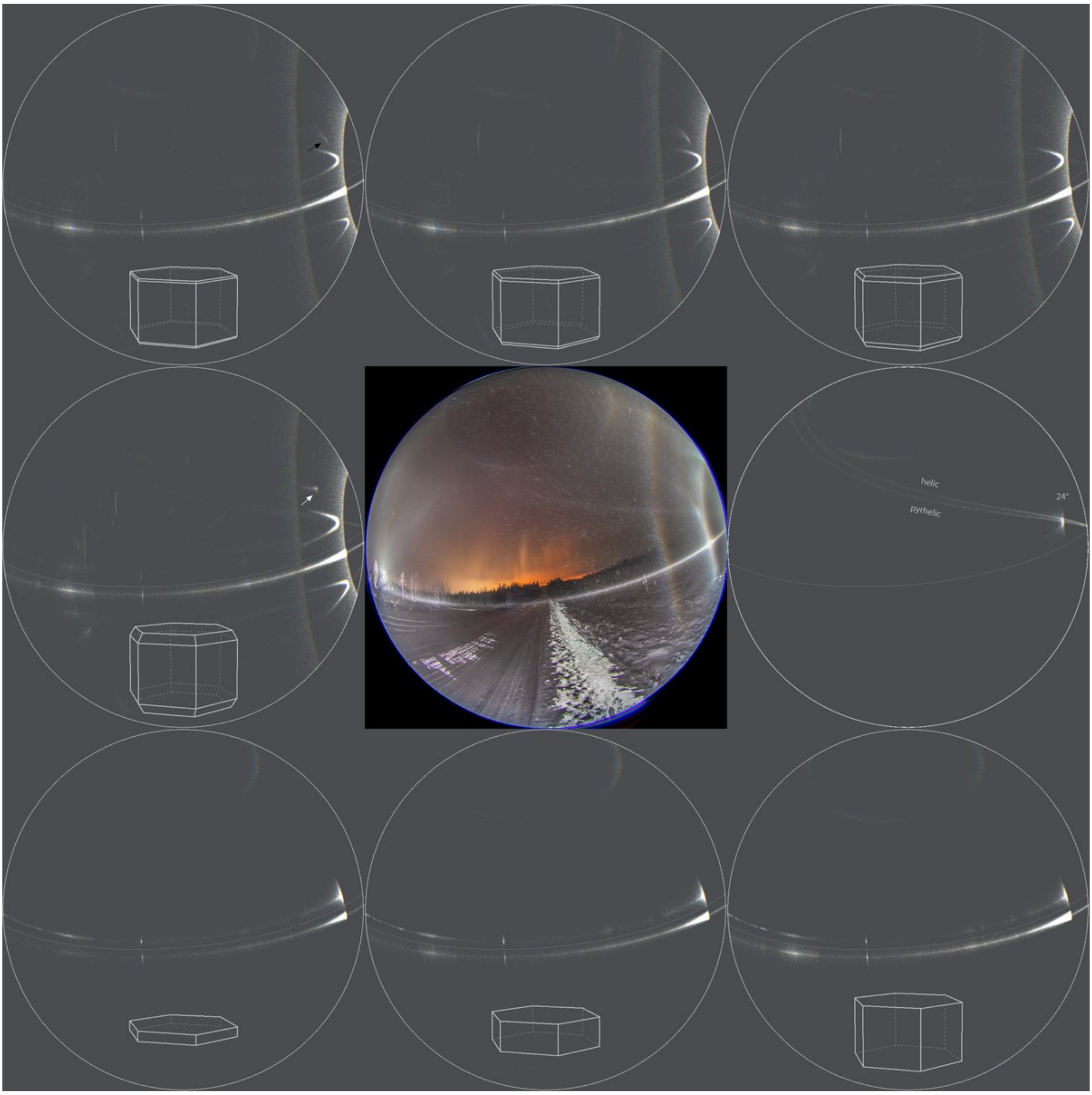
4/9 - 2024-12-25 10:35 UTC

10/11 November 2023, Rovaniemi. Another view. It is interesting that there is parhelia, countercircumnadir arc, its prism face reflection borne extension, but no counterparhelia (don't sent amateurs to halo chase: I thought all the time the upper 24° spots were counterparhelia). In the upper row I have attempted to reproduce the situation with crystals of different pyramid thicknesses. A little problem is that these scenarios produce some extra stuff that is not seen in the stack. I wouldn't worry about something missing down where the city lights are disturbing (and here is another amateur hour: the other side would of been dark), but perhaps we should see in the image at least that brightest vertical stripe higher up. Yet the simulation is wrong also in the way that 24° plate arcs are strong hooks. But then again, this is something we are often wrestling with when simulating odd radius displays.

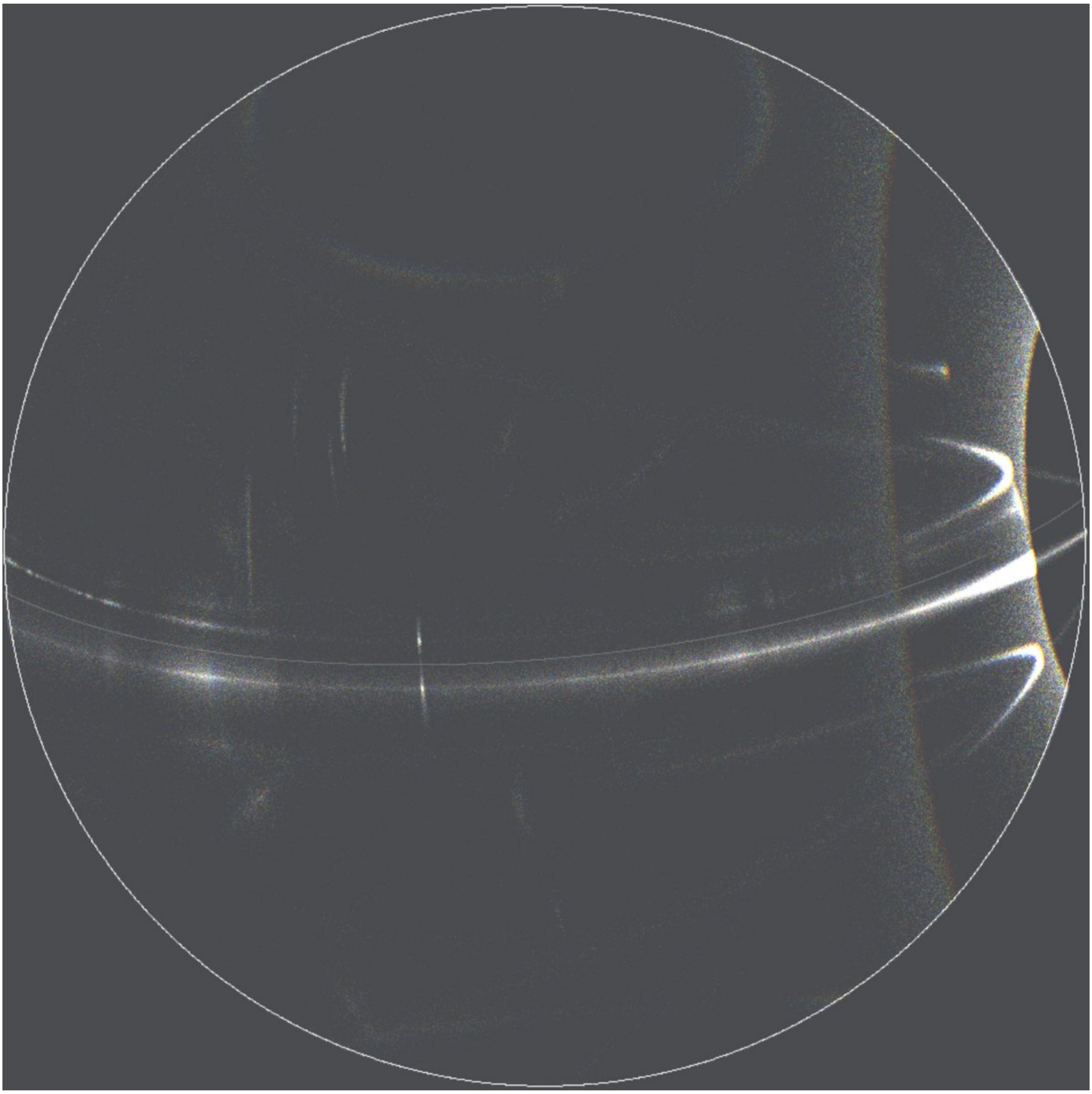
Be it as it may, the bottom row shows that what comes to normal plates, counterparhelion and countercna come with all crystal thicknesses and the thicker crystals make also the latter's extension.

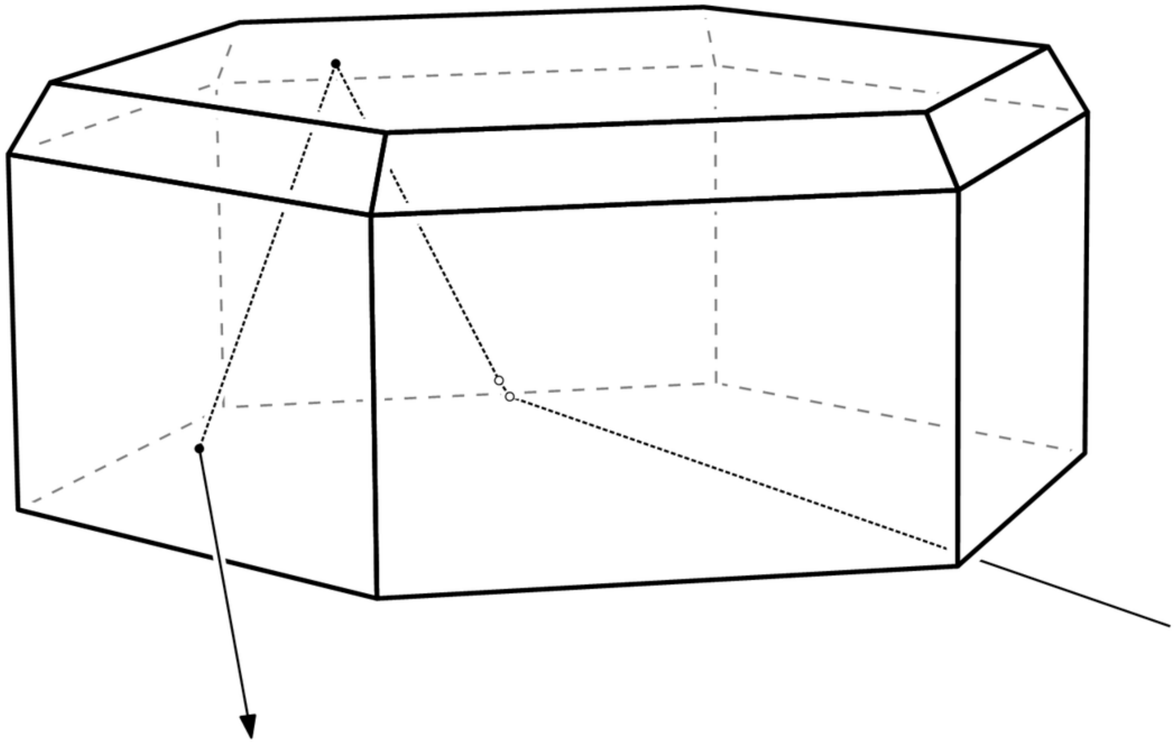
The feature pointed by the black arrow in the upper row is from 3-1-15 rays while the feature pointed by the white arrow in the middle row simulation is from 13-15. The latter is the familiar upper 35° plate arc. The former is a 24° counterplate arc, one of many, all as yet unobserved. Although it looks like 3-1-15 is better fit, in the lamp direction views – to which I will return later – my analysis rather gives the spot 13-15 identity. But as the difference between whether 13-15 or 3-1-15 is produced is only the pyramid thickness, there is really no reason why the former would be favoured over the latter. Sometimes you need to use very thin pyramid layer to simulate displays. Ice fog composition always varies and there may have been a stage here which produced 3-1-15 instead of 13-15.

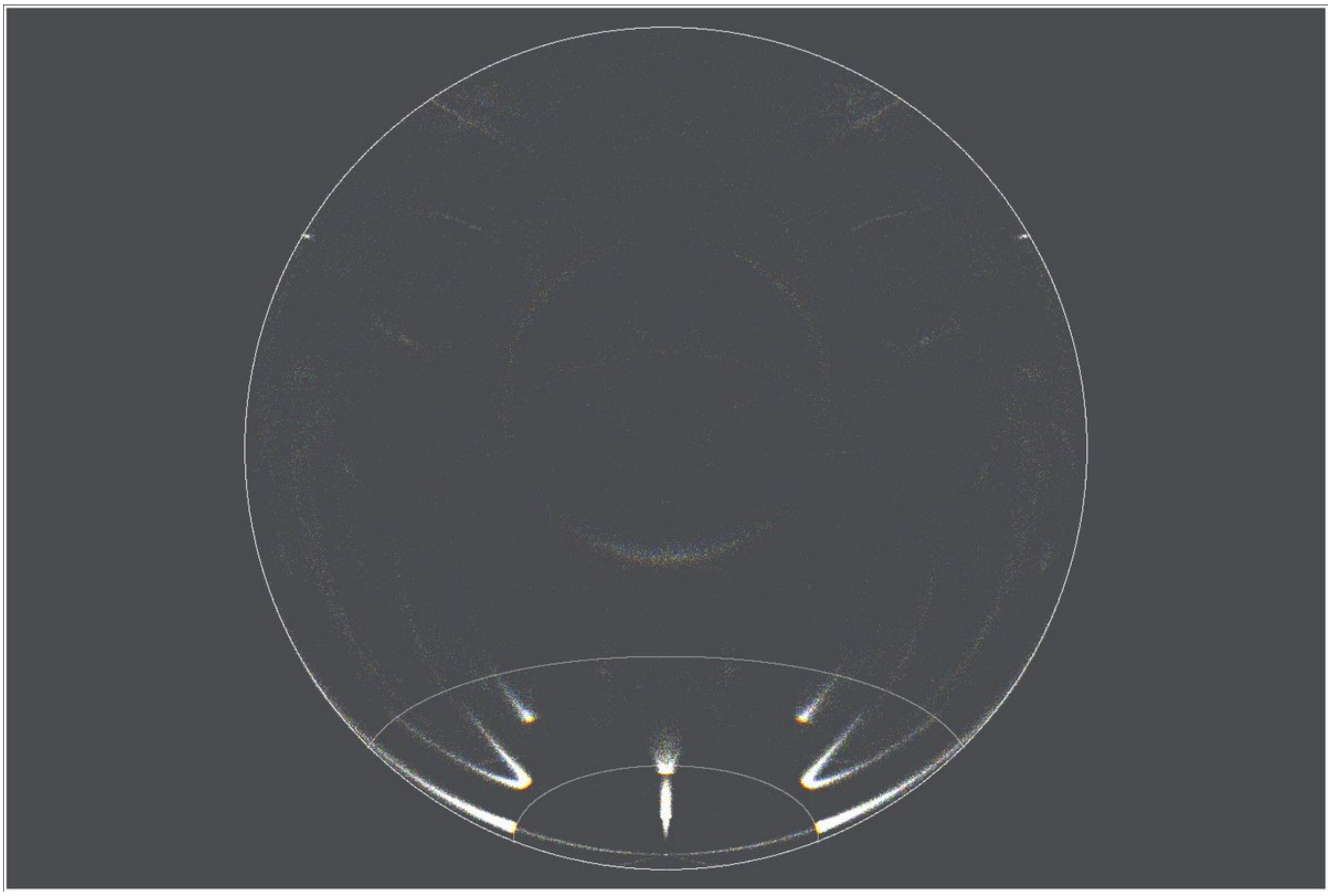
As to the helic arc, I think it is of the normal type, not the pyramid one. Counteranthelic arc is also seen so normal helic arc is expected. In the simulations I have not included column and Parry populations.



10/11 November 2023. A different simulated view of the situation in which pyramid crystal allows for parhelia, counter-cna and its extension but no counterparhelia. The responsible crystal is shown with the extension raypath. Last is one of the simulations from the upper row of the above post superposed with one of the lower row simulations to show the location of counterparhelion relative to 24° plate arc. All simulations are for the estimated lamp elevation of -3.5 degrees.







6/9 - 2024-12-26 10:14 UTC

10/11 November 2023. Checking out in this and next two posts the identity of what I have supposed are 35° plate arcs in this lamp direction view. Here the simulated spots are 35° plate arcs. Looks like they are too high. The gif is greyscale because colored was bad quality.

[video: 1872224465733767679-Gft5EAdXIAAj7sI.mp4]

7/9 - 2024-12-26 10:18 UTC

10/11 November 2023. Here the simulated spots are 35° column arcs. The location matches better but shape maybe not.

[video: 1872225452833857684-Gft6RxuXkAAyWn5.mp4]

8/9 - 2024-12-26 10:20 UTC

10/11 November 2023. Here the simulated spots are 24° counterplate arc 3-1-15. They are clearly too low.

[video: 1872225932603519457-Gft7LL-WUAAXEIl.mp4]

9/9 - 2024-12-26 10:36 UTC

10/11 November 2023. Versions of the stack. Looks like there is a suggestion of 9° column arcs but it beats me they are not clearer given how well formed 24° column arcs are. Maybe there is also lower 9° plate. Which would be of the B type 23-6 because of the low lamp.

Initially I was thinking the 24° column arc spots were the anomalous Wegener/Hastings, but

Lefaudeux set me right from that misconseption.